

PHY 521: Stars

Homework #1

Due: 2025-09-11 (by the end of the scheduled class period)

Be sure to show all work. Assignment is turned in via Brightspace assignment upload. You should make a single, legible PDF with your solutions. You must show your work and include units with all numbers, all throughout each problem.

While you are free to discuss problems with your classmates, your work must be your own. Any use of AI/LLM must be disclosed. All calculations (including any code you may write) and writing up of the problem solution must be your own work.

Unless otherwise noted, all problems carry equal weight.

1. *Stellar structure.* Consider a mass profile in a star:

$$m(r) = 4\pi R_\star^3 \beta \left[-\frac{r \cos(\pi r / R_\star)}{\pi R_\star} + \frac{\sin(\pi r / R_\star)}{\pi^2} \right]$$

where β is a constant and $R_\star$ is the radius of the star. Here, $m(r)$ is the mass enclosed inside of radius r .

- (a) What is the density profile, $\rho(r)$, that gives rise to this mass?
- (b) What is β in terms of the central density? (Note: you might need to consider the limit as $r \rightarrow 0$ and use L'Hôpital's rule)
- (c) What is the ratio of the central density to the average density, $\rho_c / \bar{\rho}$?

Note: we'll see later that this solution is related to a simplified solution for the stellar structure equations called a polytrope.

2. *Terrestrial hydrostatic equilibrium.* In class, we applied *hydrostatic equilibrium* to the atmosphere, but it also applies to the ocean. The major difference is that water, to a good approximation, is *incompressible*—you can take its density to be constant. Furthermore, we can take the acceleration of gravity to be constant, since the depth of the ocean is thin compared to the radius of the Earth.

Using this approximation, and the equation of hydrostatic equilibrium, find the depth at which the pressure in the ocean is 90 bars (the atmospheric pressure on the surface of Venus).

Side note: the reason that we can assume that water is incompressible is that it does not obey the ideal gas law, but rather a different relation where pressure is proportional to density to a high power.

3. *Virial theorem.* (this is HKT Ex. 2.6) Consider a binary pair of masses M_1 and M_2 in circular orbits about their common center of gravity. The masses are separated by a distance a . M_1 rapidly loses mass in a spherically symmetric way (in, for example, an ideal supernova explosion). The mass is lost to the system entirely. If the stars' initial velocities are v_1 and v_2 , then the velocity of M_{rem} —as what's left of M_1 as a remnant—will still be v_1 after the mass loses. Velocities and masses of the initial system are related by $M_1 v_1 = M_2 v_2$ (as simple dynamics will show). Show that the following is necessary to unbind the system:

$$\frac{M_{\text{rem}}}{M_1 + M_2} \leq \frac{1}{(1 + M_2/M_1)(2 + M_2/M_1)} < \frac{1}{2}$$

Note: that the virial theorem applies to the original binary system (this is true of any bound system of two masses).

4. *Hydrostatic equilibrium.* (based on HKT 1.2) In class we did the constant-density stellar model and argued that the central pressure we found is the minimum central pressure we would expect in a star. Here we find a different estimate of this bounds.

Starting with the pressure as:

$$\frac{dP}{dM} = -\frac{GM}{4\pi r^4}$$

integrate from the center of the star outward to the surface, using the fact that the surface pressure is zero. In order to do the integral, we need to know what $r(M)$ is. Instead take $r = R_*$ —the radius of the star, and complete the integral.

Argue that this is a lower bounds.

5. *Blackbody relations.* Recall that the Planck function in terms of wavelength is:

$$I(\lambda, T) = \frac{2hc^2/\lambda^5}{e^{hc/\lambda kT} - 1} \quad (1)$$

- (a) Derive Wien's law by setting substituting $x = \lambda kT/hc$ and finding the extrema of $I(x)$ —this will require you to find a root numerically.
- (b) Earth behaves as a blackbody—this means that the absorption of solar radiation is balanced by the thermal emission from Earth. Using this fact, the luminosity of the Sun, and the distance to the Sun, compute the expected surface temperature of Earth. Assume that clouds reflect 40% of the incident solar flux back into space, so only 60% is absorbed.